

1 Basic Numbers & Arithmetic

1.1 What Number?

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x - \sum_{k=0}^{\infty} \frac{1}{k!}$$

1.2 What Number?

$$\prod_{p \text{ prime}} \left(1 - \frac{1}{p^2}\right)^{-1} \cdot \frac{6}{\pi^2}$$

1.3 What Number? Hint: Euler's Identity variant

$$-\exp(i\pi)$$

1.4 What Number Hint: (Set Theory / Peano)

$$|\{\emptyset, \{\emptyset\}\}|$$

1.5 What Simple Operation?

$$(x \oplus y) + 2(x \wedge y)$$

1.6 What Number Are We Multiplying By?

$$\int_0^x 2 \, dt$$

1.7 What Operation?

$$\arg \min_x ((x - a)^2 + (x - b)^2)$$

1.8 What Function?

$$\frac{d}{dx} (x + \sin^2(x) + \cos^2(x)) - 1$$

1.9 We Know This Better As What?

$$\sqrt{x^2}$$

1.10 What Are We Doing Here?

$$\sum_{i=1}^N \mathbb{I}(x_i > 0) \quad \text{where } \mathbb{I}(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{otherwise} \end{cases}$$

1.11 What Are We Doing Hint: Softmax

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

1.12 What Do We Normally Call This?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

1.13 What Are We Doing To a And b?

$$\begin{bmatrix} a \\ b \end{bmatrix} \leftarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

1.14 Checking What Exactly?

$$x \pmod{2} \equiv 0$$

1.15 15. How Is This Like The Number 1? Hint: Matrices

$$\det(I_n) = 1^n$$